

Heat, Kinetic Theory and Thermodynamics, **solved**

Step-by-step solutions to every practice problem in Chapter 12. Try each problem on your own before you read its solution.

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P12.1

Metals expand more than glass when heated. Steel has $\alpha \approx 12 \times 10^{-6} \text{ K}^{-1}$, aluminium about 23×10^{-6} , and glass about 9×10^{-6} . The metal lid also conducts heat well, so it warms up quickly, while the glass rim stays cooler. Its diameter therefore grows more than the jar's mouth, which loosens the grip of the thread. A hole in a plate expands just as if it were made of the material itself, so the lid's opening gets larger.

P12.2

Both start at 100°C , but steam that touches skin first **condenses**, releasing its latent heat of $2.26 \times 10^6 \text{ J/kg}$. Only then does the resulting water cool towards skin temperature. For 1 g reaching skin at 37°C :

- Water at 100°C releases $0.001(4186)(63) = 264 \text{ J}$.
- Steam releases $2260 + 264 = 2524 \text{ J}$, almost **ten times** as much, and delivers it quickly to a small area of skin.

P12.3

Average kinetic energy: the same. At a given temperature, every molecule has the same average translational kinetic energy, $\frac{3}{2}kT$, whatever its mass.

Rms speed: different. Since $\frac{1}{2}mv_{\text{rms}}^2$ is the same for both gases,

$$\frac{v_{\text{rms,H}_2}}{v_{\text{rms,O}_2}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{H}_2}}} = \sqrt{\frac{32}{2}} = 4.$$

Hydrogen molecules move four times faster on average.

P12.4

No, the kitchen gets warmer. A refrigerator is a heat pump. It moves heat Q_C out of its interior and rejects $Q_H = Q_C + W$ from the coils at the back, where W is the electrical work.

With the door open, the “interior” is the kitchen itself, so the heat taken out at the front simply goes back in at the back. The net effect on the kitchen is the extra energy W , which ends up as heat. Energy conservation guarantees that the room warms. To cool a room, the heat must be dumped *outside* the room, which is exactly what an air conditioner does with its outdoor unit.

P12.5

The ring’s inner diameter must grow by $\Delta D = 5.000 - 4.990 = 0.010$ cm. The hole expands like the metal around it:

$$\Delta D = \alpha D_0 \Delta T \Rightarrow \Delta T = \frac{0.010}{(12 \times 10^{-6})(4.990)} = 167 \text{ K}.$$

The ring must be heated to $20 + 167 = 187$ °C, or a little more in practice for clearance. When it cools, it grips the shaft with a large thermal stress. This is how gears, bearings and railway wheels are fitted.

P12.6

Does it all melt?

- Heat needed to melt all the ice: $Q_{\text{melt}} = 0.050(3.34 \times 10^5) = 16\,700 \text{ J}.$
- Heat available from cooling the water to 0 °C: $0.300(4186)(30) = 37\,700 \text{ J}.$

Since $37\,700 > 16\,700$, all of the ice melts.

Final temperature. Heat gained by the ice (melting, then warming from 0 to T) equals heat lost by the warm water:

$$16\,700 + 0.050(4186)T = 0.300(4186)(30 - T).$$

$$16\,700 + 209T = 37\,670 - 1256T \Rightarrow 1465T = 20\,970 \Rightarrow T = 14.3\,^{\circ}\text{C}.$$

P12.7

(a) Extra internal energy:

$$Q = mc\Delta T = 70(3500)(2.0) = 4.9 \times 10^5 \text{ J}.$$

That is about 117 kcal.

(b) Time:

$$t = \frac{4.9 \times 10^5}{50} = 9.8 \times 10^3 \text{ s} \approx 2.7 \text{ h}.$$

In practice the body speeds this up by shivering, which can raise heat production several-fold, and by narrowing the blood vessels in the skin to reduce heat loss. These are the “chills” at the start of a fever.

P12.8

Thermal resistances (per unit area, L/k):

- Brick: $0.20/0.70 = 0.286 \text{ m}^2\text{K/W}$.
- Insulation: $0.10/0.040 = 2.50 \text{ m}^2\text{K/W}$.
- Total: $2.786 \text{ m}^2\text{K/W}$.

(a) Heat loss:

$$P = \frac{A\Delta T}{\sum L/k} = \frac{20(25)}{2.786} = 180 \text{ W}.$$

(b) Brick alone:

$$P = \frac{20(25)}{0.286} = 1.75 \text{ kW}.$$

The insulation cuts the heat loss by a factor of about 10.

(c) **Boundary temperature.** The same heat flux passes through each layer, so the temperature drop across each layer is proportional to its resistance. The insulation, on the inside, takes

$2.50/2.786 = 89.7\%$ of the 25 K drop, which is 22.4 K. So the boundary is at

$$20 - 22.4 = -2.4\text{ }^{\circ}\text{C}.$$

The brick then takes the remaining 2.6 K. Note that the boundary is below freezing, so moisture that seeps through the insulation can condense and freeze there. This is why builders put a vapour barrier on the warm side.

P12.9

(a) **Temperature**, from Wien's law:

$$T = \frac{2.898 \times 10^{-3}}{502 \times 10^{-9}} = 5.77 \times 10^3\text{ K}.$$

(b) **Power per square metre**:

$$\sigma T^4 = (5.67 \times 10^{-8})(5770)^4 = 5.67 \times 10^{-8} \times 1.11 \times 10^{15} = 6.3 \times 10^7\text{ W/m}^2.$$

(c) **Total power**:

$$P = 4\pi R^2 \sigma T^4 = 4\pi(6.96 \times 10^8)^2(6.3 \times 10^7) = 3.8 \times 10^{26}\text{ W}.$$

This agrees well with the measured solar luminosity, $3.83 \times 10^{26}\text{ W}$.

P12.10

At constant volume, the *absolute* pressure is proportional to the absolute temperature.

- Initial absolute pressure: $200 + 101.3 = 301.3\text{ kPa}$, at $T_1 = 283\text{ K}$.
- Final absolute pressure:

$$p_2 = 301.3 \times \frac{313}{283} = 333.2\text{ kPa}.$$

- Final gauge pressure: $333.2 - 101.3 = 232\text{ kPa}$.

So the gauge reading rises by 16%. That is why tyre pressures should be checked when the tyres are cold. A common mistake is to use gauge pressure, or temperatures in $^{\circ}\text{C}$, in the gas law. Both give wrong answers.

P12.11

$$v_{\text{rms}}(\text{H}_2) = \sqrt{\frac{3(8.314)(300)}{0.002}} = 1.93 \text{ km/s}, \quad v_{\text{rms}}(\text{O}_2) = \sqrt{\frac{3(8.314)(300)}{0.032}} = 0.48 \text{ km/s}.$$

Both are well below the escape speed of 11.2 km/s. But molecular speeds are *distributed*: some molecules move several times faster than v_{rms} . At the top of the atmosphere, where the temperature can exceed 1000 K and collisions are rare, a small but steady fraction of hydrogen molecules exceed escape speed and leak away into space. Over billions of years this has removed almost all of the Earth's free hydrogen and most of its helium. The rule of thumb is that a planet keeps a gas for geological time only if v_{rms} is below about one-sixth of the escape speed. Oxygen and nitrogen pass easily, but hydrogen does not.

P12.12

(a) **Work:**

$$W = p \Delta V = (1.0 \times 10^5)(3.0 \times 10^{-3}) = 300 \text{ J}.$$

(b) **Internal energy.** For a diatomic gas, $\Delta U = \frac{5}{2}nR\Delta T = \frac{5}{2}p\Delta V$, using $pV = nRT$ at constant p :

$$\Delta U = \frac{5}{2}(300) = 750 \text{ J}.$$

(c) **Heat, from the first law:**

$$Q = \Delta U + W = 750 + 300 = 1050 \text{ J}.$$

Check: $Q = nC_p\Delta T = \frac{7}{2}p\Delta V = 1050 \text{ J}.$ ✓

P12.13

Label the states. State 1 is at volume V_1 and temperature T_1 , before compression. Compression takes it to state 2 at (V_2, T_2) . Combustion at constant volume takes it to state 3 at (V_2, T_3) . Expansion takes it to state 4 at (V_1, T_4) . Exhaust at constant volume returns it to state 1.

Heat. Heat enters only during combustion and leaves only during exhaust, both at constant volume:

$$Q_H = nC_V(T_3 - T_2), \quad Q_C = nC_V(T_4 - T_1).$$

Adiabats. Using $TV^{\gamma-1} = \text{constant}$ on each adiabat:

$$T_2 = T_1 r^{\gamma-1}, \quad T_3 = T_4 r^{\gamma-1}.$$

Subtracting: $T_3 - T_2 = (T_4 - T_1)r^{\gamma-1}$.

Efficiency:

$$\eta = 1 - \frac{Q_C}{Q_H} = 1 - \frac{T_4 - T_1}{(T_4 - T_1)r^{\gamma-1}} = 1 - r^{1-\gamma}. \checkmark$$

Numbers:

- $r = 10$: $\eta = 1 - 10^{-0.4} = 1 - 0.398 = 60\%$.
- $r = 18$: $\eta = 1 - 18^{-0.4} = 1 - 0.315 = 69\%$. (A real diesel cycle has constant-pressure combustion, which gives slightly less, but the trend holds.)

Why real engines manage only about 30–45%. Heat leaks to the cylinder walls and coolant. There is friction. Combustion takes a finite time rather than being instantaneous. The working gas is not an ideal gas with constant γ . Energy is spent pumping gas in and out. A higher compression ratio raises the efficiency, but in a petrol engine it is limited by premature ignition (“knock”). Diesels avoid this by injecting the fuel only after compression.

P12.14

Data. $V_1/V_2 = 10$, $T_1 = 300$ K, $p_1 = 1.0$ atm, $n = 1$ mol.

(a) **Isothermal compression:**

- $T_2 = 300$ K.
- $p_2 = 10$ atm.
- Work done on the gas: $W_{\text{on}} = nRT \ln 10 = 8.314(300)(2.303) = 5.7 \times 10^3$ J.

(b) **Adiabatic compression:**

- $T_2 = 300(10)^{0.4} = 754$ K.
- $p_2 = 1.0(10)^{1.4} = 25$ atm.
- Work done on the gas, which all becomes internal energy since $Q = 0$:

$$W_{\text{on}} = \Delta U = nC_V \Delta T = \frac{5}{2}(8.314)(454) = 9.4 \times 10^3 \text{ J.}$$

Why more work? In the adiabatic compression, the gas heats up as it is squeezed, so its pressure climbs faster than $1/V$. The force on the piston is therefore larger throughout the stroke, and so is the area under the pV curve. In the isothermal case, the energy put in leaves continuously as heat, so the pressure rises only as $1/V$. Engineers exploit this with *intercoolers* on air compressors and turbochargers: cooling the gas between stages makes the compression closer to isothermal and saves work.

P12.15

Find k from the two measurements, one hour apart:

$$\frac{30.0 - 20}{28.0 - 20} = e^{k(1 \text{ h})} \Rightarrow k = \ln \frac{10}{8} = 0.223 \text{ h}^{-1}.$$

Time from 37.0 °C to 30.0 °C:

$$t = \frac{1}{k} \ln \frac{37.0 - 20}{30.0 - 20} = \frac{\ln 1.7}{0.223} = \frac{0.531}{0.223} = 2.4 \text{ h}.$$

So death occurred about 2.4 h before 10:00, around 7:40.

Limitations.

- The core temperature often stays steady for an hour or so after death (the “plateau”), before it starts to fall.
- Body size, fat, clothing, posture, air currents and humidity all change k .
- The body may not have been at 37.0 °C at death, because of fever or hypothermia.
- The room temperature may have changed over time.

Forensic pathologists use corrected cooling curves, such as Henssge’s nomogram, and quote an uncertainty of a few hours. Even so, the physics is the basis of the method.

P12.16

Mass of air breathed per second:

$$\dot{m} = \frac{7.5 \text{ L/min} \times 1.29 \text{ g/L}}{60} = 0.161 \text{ g/s} = 1.61 \times 10^{-4} \text{ kg/s}.$$

(a) **Warming the air** from 0 °C to 37 °C:

$$P_{\text{warm}} = \dot{m}c \Delta T = (1.61 \times 10^{-4})(1005)(37) = 6.0 \text{ W}.$$

(b) **Humidifying it.** The water evaporated per second is

$$\frac{7.5 \text{ L/min} \times 44 \text{ mg/L}}{60} = 5.5 \text{ mg/s} = 5.5 \times 10^{-6} \text{ kg/s},$$

so

$$P_{\text{humid}} = (5.5 \times 10^{-6})(2.4 \times 10^6) = 13 \text{ W}.$$

Total: about 19 W, roughly 20% of the resting metabolic rate. Evaporation costs twice as much as warming the air. This is one reason why breathing cold, dry air is so dehydrating and tiring at high altitude, and why mountaineers and Arctic workers sometimes use heat-and-moisture-exchanging face masks.

P12.17

(a) **Stress and force.**

$$\sigma = Y\alpha \Delta T = (2.0 \times 10^{11})(12 \times 10^{-6})(35) = 84 \text{ MPa}.$$

$$F = \sigma A = (8.4 \times 10^7)(7.5 \times 10^{-3}) = 6.3 \times 10^5 \text{ N}.$$

That is about 630 kN, the weight of some 64 tonnes.

(b) **Buckling.** A long, slender rod under a large compressive force is unstable. Like a ruler squeezed from both ends, it can relieve the compression by bending sideways. Above a critical force, which depends on how well the rail is restrained by the sleepers and ballast, the track suddenly kinks sideways, which can derail a train.

Prevention:

- The rail is laid and welded at a chosen *stress-free temperature*, typically near the hot end of the local range, so that summer compression stays small. In winter the rail is in tension instead, which steel tolerates well.
- Heavy concrete sleepers, full ballast shoulders and strong fastenings resist sideways movement.
- Speed restrictions are imposed during heatwaves.

P12.18

Temperatures: $T_C = 255$ K and $T_H = 298$ K.

(a) Minimum power:

$$K_{\max} = \frac{T_C}{T_H - T_C} = \frac{255}{43} = 5.93, \quad W_{\min} = \frac{Q_C}{K_{\max}} = \frac{200}{5.93} = 34 \text{ W}.$$

(b) Actual power and heat rejected. With $K = 2.5$:

$$W = \frac{200}{2.5} = 80 \text{ W}, \quad Q_H = Q_C + W = 200 + 80 = 280 \text{ W}.$$

(c) Entropy rates:

$$\dot{S}_{\text{interior}} = -\frac{200}{255} = -0.784 \text{ W/K}, \quad \dot{S}_{\text{kitchen}} = +\frac{280}{298} = +0.940 \text{ W/K}.$$

$$\dot{S}_{\text{universe}} = +0.155 \text{ W/K} > 0. \checkmark$$

The interior's entropy decreases, but the kitchen's increases by more, so the second law is satisfied. For an ideal (Carnot) freezer, with $W = 34$ W and $Q_H = 234$ W, the kitchen's gain would be $234/298 = 0.784$ W/K, exactly cancelling the interior's loss, so $\dot{S}_{\text{universe}} = 0$. The extra entropy in the real case measures its irreversibility.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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